

Microeconomic Theory

Final Exam

2024

Instructions: Answering the questions in this exam is an individual effort. The estimated answering time is provided at the beginning of each question. Please justify your answers briefly.

1. (Approximately 20 minutes)

Consider the symmetric independent private values model (SIPVM) with a single bidder whose valuation x is drawn from $[0, 1]$ according to the distribution function F assumed to have a continuous and strictly positive density function f .

In each case below, indicate whether the function provided corresponds to the bidder's utility for some direct revelation mechanism (DRM). Provide the DRM or show that it does not exist.

(a)

$$(\forall x \in [0, 1]) \quad U(x) = \frac{1}{2}x^2.$$

(b)

$$(\forall x \in [0, 1]) \quad U(x) = \frac{2}{3}x^2.$$

(c)

$$U(x) = \begin{cases} 0 & \text{if } x \in [0, 1/8] \\ \frac{x}{3} - \frac{1}{24} & \text{if } x \in [1/8, 1/2] \\ \frac{x}{2} - \frac{1}{8} & \text{if } x \in [1/2, 3/4] \\ x - \frac{1}{2} & \text{if } x \in [3/4, 1] \end{cases}$$

2. (Approximately 20 minutes)

Consider the symmetric independent private values model (SIPVM) with a single bidder whose valuation x is drawn from $[0, 1]$ according to the distribution function F assumed to have a continuous and strictly positive density function f .

In each case below, indicate whether the allocation function provided corresponds to a DRM. Provide the DRM and the bidder's utility or show that it does not exist.

(a)

$$q(x) = \begin{cases} 0 & \text{if } x \in [0, 1/2) \\ 1 & \text{if } x \in [1/2, 1] \end{cases}$$

(b)

$$q(x) = \begin{cases} 0 & \text{if } x \in [0, 1/3) \\ \frac{1}{3} & \text{if } x \in [1/3, 2/3) \\ \frac{2}{3} & \text{if } x \in [2/3, 1] \end{cases}$$

(c)

$$q(x) = \begin{cases} 0 & \text{if } x \in [0, 1/4) \\ \frac{1}{4} & \text{if } x \in [1/4, 1/2) \\ \frac{3}{4} & \text{if } x \in [1/2, 3/4) \\ \frac{2}{4} & \text{if } x \in [3/4, 1] \end{cases}$$

3. (Approximately 70 minutes)

There are n bidders participating in a first-price auction. Each bidder's value is independently drawn from $[0, 1]$ according to the distribution function F assumed to have a continuous and strictly positive density f .

If a bidder's value is v and she wins the object with a bid of $b < v$; then her von Neumann-Morgenstern utility is $(v - b)^{1/\alpha}$; where $\alpha \geq 1$ is fixed and common to all bidders. Consequently, the bidders are risk averse when $\alpha > 1$ and risk neutral when $\alpha = 1$.

Given the risk aversion parameter α , let $b_\alpha(v)$ be the symmetric equilibrium bid of a bidder when his value is v .

- (a) Let $U(b_\alpha(x), v)$ denote a bidder's expected utility from bidding $b_\alpha(x)$ when her valuation is v , assuming that all other bidders employ the strategy $b_\alpha(\cdot)$. Show that

$$U(b_\alpha(x), v) = F^{N-1}(x)(v - b_\alpha(x))^{1/\alpha}$$

and that this expression is maximized when $x = v$.

- (b) Show that

$$[U(b_\alpha(x), v)]^\alpha = F^{\alpha(N-1)}(x)(v - b_\alpha(x))$$

and that this expression is also maximized when $x = v$.

- (c) Show that the equilibrium strategy is

$$b_\alpha(v) = v - \int_0^v \left(\frac{F(x)}{F(v)} \right)^{\alpha(N-1)} dx$$

$$\frac{d}{dv} [U(b_\alpha(v), v)]^\alpha = \frac{d}{dv} [F^{\alpha(N-1)}(v)(v - b_\alpha(v))] = \alpha F^{\alpha(N-1)}(v) - F^{\alpha(N-1)}(v) = (\alpha - 1) F^{\alpha(N-1)}(v) \geq 0$$